



CS 03500 - Foundations of Cybersecurity

Cyberwarfare: Information Operations in a Connected World Projects

Project Part 3:

Ensuring Defense in Depth

Name: Devon Jacob

Instructor: Michael Davie

Date: 11/18/2024

- a. The vulnerabilities of at least two defense-in-depth layers that could enable attackers to access computers on the network.

- i. Network Security – Segmentation**

Segmentation breaks the network into smaller, more manageable sections, allowing for security policies and controls to be customized for each segment. In the event of a breach, segmentation helps contain the attack to a single area, reducing its overall impact on the network (Swisher). Many power grid networks lack adequate segmentation, therefore attackers who breach the perimeter can easily navigate throughout the network, gaining access to critical system. This was evident in Russia's attack on Ukraine power grid in 2015 where attackers exploited poor network segmentation and gained access to the operational technology (Antoniuk). Antoniuk further stated that effective network segmentation between IT and OT systems and networks is a vital security measure in such environments. Implementing strong network segmentation in power grid systems is essential to limit the spread of attacks, protect critical infrastructure, and minimize potential damage from cyber threats.

- ii. Endpoint Security – Patch Management**

Patch management ensures endpoint operating systems and applications are current and mitigate common vulnerabilities and exposures (CyberArk). It is essential to regularly update endpoint operating systems and applications to ensure they are protected from known security risks. This proactive approach helps strengthen the overall security posture of the network, preventing

unauthorized access, data breaches, or system compromises that can result from outdated software. Attackers can exploit these vulnerabilities to gain remote access or execute malicious code which was evident in the WannCry ransomware attack that targeted unpatched windows systems (*What Was the WannaCry Ransomware Attack? | Cloudflare*). Regular and effective patch management is crucial in maintaining a secure and resilient network, minimizing the risk of exploitation and ensuring the protection of critical infrastructure from evolving cyber threats.

- b. The NSA's defense-in-depth strategy encompasses people, technology, and operations, offering guidelines and best practices to safeguard physical infrastructure, organizational procedures, and IT systems (CyberArk).

- i. People – Access Control**

Enforce role-based access control (RBAC) to the power grid to ensure users only access systems necessary for their duties and require multi-factor authentication (MFA) for accessing critical systems, such as Supervisory Control and Data Acquisition (SCADA) networks. This applies to the power grid computer network because it limits unauthorized access. By implementing these security measures, the power grid protection against cyber threats can be significantly enhanced, safeguarding critical infrastructure from potential vulnerabilities.

- ii. Technology – Intrusion Detection and Prevention Systems (IDPS)**

An intrusion detection and prevention system (IDPS) is a tool that monitors network activity for potential threats and takes measures to prevent any detected threats from causing harm (*What Is an Intrusion Detection and Prevention System (IDPS)?*). It applies to the power grid since it would provide early warning of potential breaches and enable rapid response. Implementing an IDPS on the power grid ensures timely detection and mitigation of cyber threats, enhancing the overall security posture and protecting vital infrastructure from attacks.

iii. Operations – Data Recovery

Data recovery is essential for a strong cyber resilience strategy following a cyber-attack, helping businesses restore operational continuity. It enables organizations to swiftly recover critical data, reduce downtime, and protect against future threats (Ray). In relation to the power grid, it would minimize operational downtime and ensure business continuity. By incorporating robust data recovery protocols, the power grid can maintain seamless operations and quickly recover from disruptions, ensuring long-term stability and resilience against future cyber threats.

- c. The use of Advanced Encryption Standard (AES) cryptographic system is a highly secure encryption technique. It uses a symmetric encryption algorithm with a block cipher, processing data in fixed-size blocks one at a time (Mujeeb). Encrypt communication between grid components, for example substations and control centers using AES-256 to prevent data interception. By deploying AES-

256 encryption for communication between grid components, the power grid can safeguard sensitive data from unauthorized access and ensure the integrity and confidentiality of critical information across the network.

Works Cited

- Antoniuk, Daryna. *Ukraine Energy Facility Took Unique Sandworm Hit on Day of Missile Strikes*. 9 Nov. 2023, therecord.media/sandworm-attack-ukraine-energy-facility-missile-strikes.
- CyberArk. "What Is Defense-in-Depth?" *CyberArk*, 23 Apr. 2024, www.cyberark.com/what-is/defense-in-depth.
- Mujeeb, Abdul. "4 Cryptographic Techniques Used in Cybersecurity." *4 Cryptographic Techniques Used in Cybersecurity*, 24 May 2024, www.quickstart.com/information-security/4-cryptographic-techniques-used-in-cyber-security.
- Ray, Kaushik. "CloudTweaks | The Role of Data Recovery in Cyber Resilience." *CloudTweaks*, 18 Nov. 2024, cloudtweaks.com/2024/11/role-data-recovery-cyber-resilience.
- Swisher, Jen. "What is Defense in Depth (DiD) in Cybersecurity." *Jetpack*, 3 Sept. 2024, jetpack.com/blog/defense-in-depth.
- What is an intrusion detection and prevention system (IDPS)?*
www.redhat.com/en/topics/security/what-is-an-IDPS.
- What was the WannaCry ransomware attack? | Cloudflare*.
www.cloudflare.com/learning/security/ransomware/wannacry-ransomware.